

GenomeIDP Developer Handover Document

Website-Wide

CSS and Responsiveness

- The current CSS is fragmented and inconsistent across pages. The main issues are:
 - Lack of responsive design – the website does not adapt well to small screens (phones/tablets). Media queries and flexible layout containers are needed.
 - UI uniformity – elements like buttons, font styles, and navigation components vary across the app. For example, multiple variants of the TopNav exist; these should be merged into one reusable component.
 - Recommend creating a central style guide and reusable components library (e.g., `components/common/` for buttons, navbars, etc.)

Deployment

- I have not started working on deployment. Two major concerns remain:
 1. R Script Execution: The backend depends on running an R script to cluster users with experts. The new developer will need to install the correct version of R and ensure the `child_process` spawn logic works on the deployment server (tested locally, not in production).
 2. MongoDB Integration: Ensure MongoDB URI is correctly set via environment variables and linked to the backend via Mongoose. The DB schema (`Answer.js`) is already built.

Page 1 (Landing Page)

- The email prompt at the bottom is a placeholder only. Backend API and logic need to be added (e.g., store user emails for newsletter or updates if opted in).
 - The intro animations (slide-in/fade effects) can be made smoother using `framer-motion` or CSS transitions. Currently, performance is slightly choppy.
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Page 2 (Information Page)

- The images are high resolution, which delays loading times. Recommend compressing using a tool like TinyPNG or exporting optimized versions from source files.
 - Lazy loading can be implemented with `loading="lazy"` tags or libraries like `react-lazyload`.
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Login / Signup Pages

- The eye icon used to toggle password visibility is not responsive; on small screens, it drifts outside the input box. This needs to be realigned using relative positioning inside the input group.
 - There is no “Forgot Password” functionality. Since we wanted to preserve anonymity, consider:
 - Allowing users to select a security question at signup and verifying this instead of sending a reset email.
 - Alternatively, consider optional email entry only for password recovery.
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Form Page (Survey)

- Create a way to lock switching to the next section using the side navbar unless the section has a) has been completed already or b) is the next one in line to be completed. We want to force users to finish a section before moving on.
- The answers are stored in MongoDB using the `Answer` schema, and token-based auth is used to identify sessions.

- Fix sidenav CSS, currently, when a section is hovered, its outline is outside the sidenav boundaries.

Form Schema Description

The survey form data is stored in a MongoDB collection using the following schema structure:

Section Schema (`sectionAnswerSchema`)

- Represents a specific survey section (e.g., "Skills", "Interests").
 - Fields:
 - `section`: A string from a predefined list of valid sections. Enforced using `enum`.
 - `answers`: A Map object where each key is a question identifier and each value is a numeric rating from 1 to 5.
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Results Page

Side Nav

- Needs CSS rework: current implementation has issues with sticky positioning and padding.
- Consider turning it into a reusable component as it will likely appear in future pages.

Values Tab

- Currently contains placeholder content.
- Next steps:
 - Load user answers from backend (`/api/form/`).
 - Filter for values with a rating of 5/5.
 - Display the matched value prompts using the static JSON `valuePrompts.json`.
 - Map question IDs to prompts and display them as a table or list.

Careers Page

File Name Matching Bug

- Some career names with slashes ("/") cause errors because file names cannot contain slashes. Fix this by:
 - Replacing "/" with an underscore or using a safe `encodeURIComponent` method when loading the JSON files.
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- I use the same table from the career matches everywhere else in the website, remove the hover effect from tables where the rows are not clickable links.
- Test matching algorithm on a sample of low user answers (all 1s). The algorithm has only been tested with high user answers (all 5s).

Navigation Button

- There needs to be a "Back to Career Matches" button for users to return from the career details view to the main matches list.

CSS Fixes

- The Career Matches and Career Resources pages need aesthetic and responsive layout improvements.
 - Add consistent background color, margin/padding cleanup, and mobile-friendly grid layouts.
 - For CareerResourcesPage, a Back button was added but can be improved with icons and hover states.

Missing Features (To Be Built)

Progress Tracking / Journaling

- Erin requested features that allow users to:
 - Track career planning steps (checklists, milestone tables).
 - Add journal entries, store them in MongoDB, and access/edit later.

- These pages are detailed in the original requirements document.
- These pages are not implemented yet, but the backend structure supports this.

Certificate Generator

- On survey completion, Erin wanted a downloadable certificate.
 - Since accounts are anonymous, the user should be prompted for a name only if they want to generate a certificate.
 - Use libraries like `html2canvas` or `jspdf` to create printable/downloadable certificates.
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General Notes

- JWT Tokens do not expire. You can update the expiry time in the auth middleware (`authMiddleware.js`) if needed.
- Survey Question Mismatch:
 - There's a discrepancy between the requirements doc and scorer survey responses. I've aligned the app to follow the scorer survey for consistency in matching.
 - Total questions used in scoring = 171 (not 175). Values questions are excluded from the algorithm.

SKILLS

- Missing in doc:
 - "Broad knowledge of how drugs / biologics / devices are regulated..." → Add this to the website.
- Out of order (practical skills):
 - Order should be:
 1. Establishing frameworks to collect, analyze, and interpret data

2. Tolerance / acceptance of uncertainty and ambiguity

3. Tolerance for repetitive / inconclusive work

- **Missing in scorer form:**
 - **Communicating how individual findings fit into broader program**
 - **Creating communications plans**
 - **Effective peer communication**
 - **Navigating difficult conversations**
- **Reorganization:**
 - **Some questions were moved between "Communication" and "Professionalism" categories. The scorer sheet structure is preferred.**

INTERESTS

- **Misplaced:**
 - **"Lifelong learning..." is in Discipline-specific but should be in Professional Management.**
 - **Duplicate/split:**
 - **Should be one: "Writing variant analysis reports based on professional guidelines and lab/clinic requirements"**
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Matching Algorithm Notes

- **The algorithm is based on nearest neighbor comparison between user answers and expert profiles across SKILLS + INTERESTS only.**
- **Matching logic is handled in R and returns a profession-count object.**
- **Frontend parses this into ranked matches (rank 1 = highest count) and omits unmatched careers.**

- Alignment scores, low-variance questions, and dotplots are rendered from JSON inputs in [/public/data](#).

Let me know if you need any clarification. Good luck with continuing GenomeIDP!

Faizan Hasan